Fourier Series and Fourier Transform - Notes

Fourier Series - Definition: The Fourier Series represents a periodic signal as a sum of sine and cosine waves or complex exponentials.

Key Points:
- Applicable to periodic signals
- Produces discrete frequency components
- Represents signal as a sum of harmonics

Applications of Fourier Series:
1. Signal analysis of periodic signals
2. Communication systems (AM/FM)
3. Electrical circuit analysis
4. Music and sound analysis
5. Mechanical vibration analysis

Fourier Transform - Definition: The Fourier Transform converts a non-periodic signal into its frequency components.

Key Points:
- Applicable to non-periodic signals
- Produces continuous frequency spectrum
- Uses integral representation

Applications of Fourier Transform:
1. Image processing (compression, filtering)
2. Audio processing (noise reduction)
3. Wireless communication (OFDM)
4. Medical imaging (MRI)
5. Seismology (earthquake analysis)
6. Data science and machine learning

Differences:
Fourier Series: periodic signals, discrete frequencies
Fourier Transform: non-periodic signals, continuous spectrum